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# Exploratory clinical and immunological changes following Tixagevimab/Cilgavimab administration in immunodeficient patients with SARS-CoV-2-associated ME/CFS and persistent circulating spike protein

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ABSTRACT

*Background:* Persistent circulating SARS-CoV-2 spike protein has been proposed as a potential contributor to long-COVID pathophysiology. Whether administration of spike-targeting monoclonal antibodies is followed by changes in spike protein concentrations, autonomic receptor autoantibodies, or clinical outcomes remains uncertain.

*Methods:* In this prospective, non-randomized observational study, 21 participants were enrolled. Ten immunodeficient patients with long-COVID-associated ME/CFS and detectable circulating spike protein received Tixagevimab/Cilgavimab for SARS-CoV-2 pre-exposure prophylaxis. One patient subsequently withdrew consent, leaving nine patients for paired baseline and 30-day follow-up analyses. Eleven healthy individuals with previous SARS-CoV-2 infection but without long COVID provided descriptive reference data and were not considered an efficacy control group.

*Results:* β1-adrenergic receptor autoantibody concentrations decreased from 26.8 ± 18.2 to 21.2 ± 14.7 U/ml (unadjusted $p = 0.003$). No statistically significant changes were observed for β2-adrenergic or M3/M4-muscarinic receptor autoantibodies. Median spike protein concentrations changed from 63 pg/ml (IQR 9–127) to 26 pg/ml (IQR 10–120; $p = 0.63$), with individual decreases observed in five of nine patients. None of the assessed clinical outcomes changed significantly at group level. Changes in spike protein concentrations were not significantly correlated with changes in the assessed autonomic receptor autoantibodies.

*Conclusion:* Tixagevimab/Cilgavimab administration was followed by a nominally significant within-patient reduction in β1-adrenergic receptor autoantibody concentrations, but not by conclusive changes in circulating spike protein or clinical outcomes. These findings are preliminary and hypothesis-generating. The small sample size, non-randomized design, brief 30-day follow-up, inclusion of only immunodeficient patients, and absence of an untreated disease-matched control group preclude conclusions regarding therapeutic efficacy.

## 1. Introduction

Two years after the pandemic, the prevalence of long-COVID syndrome is still high with enormous effects on both individual health and socioeconomic costs. A recent study described a long-COVID prevalence for those who were vaccinated of 5.3% and unvaccinated of 9.5% [1]. Long-COVID refers to a range of symptoms persisting or emerging three months after an initial SARS-CoV-2 infection, lasting at least two months without other explanations, as defined by the WHO [2]. This condition can impact nearly every organ system, causing respiratory, neurological, cardiovascular, gastrointestinal, and mental health issues. Within other common symptoms, a significant proportion of patients with long-COVID demonstrates a combination of neurological and cognitive symptoms known as myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS). ME/CFS has been documented since 1988 and occurs mostly following infections. Though its pathophysiology is not fully understood, pathogen (viral) persistence, inflammation and autoimmunity have been considered as proposed mechanisms in virus-associated ME/CFS including long-COVID associated ME/CFS. In line with this proposed pathophysiology, we and other demonstrated an

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association between the concentrations of autoantibodies directed against receptors of autonomic nervous systems such as beta-1 and −2 adrenergic and M3 and M4 cholinergic receptor autoantibodies and the severity of neurological symptoms in long-COVID with ME/CFS [3]. Furthermore, a recent prospective trial showed an association between long-COVID, brain injury markers, gray matter volume and global cognitive impairment, describing objectifiable neurologic impairments [4]. To date, there is no evidence-based therapy for this entity. Trying to address the proposed role of autoantibodies in ME/CFS development, we and another group removed autoantibody by immunoadsorption with a limited clinical success [5,6].

There is evidence suggesting virus persistence in patients with long-COVID. Thus, persistent SARS-CoV-2 proteins and mRNA have been detected in various tissues, e.g. in lungs, lymph nodes, cerebrospinal fluid and plasma [7–11]. Furthermore, there is evidence supporting persistence of the soluble spike protein in plasma of patients. Interestingly, persistence of soluble spike protein is associated with symptoms of long-COVID [12]. It may be hypothesized that persisting spike protein is associated with the perpetuation of autoimmune inflammation.

Addressing a possible pathophysiological role of spike protein persistence in long-COVID, Scheppke et al. infused spike-specific monoclonal antibodies (mAb) in three cases of severe long-COVID and demonstrated rapid and complete remission in all patients. This observation called for further studies that will analyze targeted effects of spike-specific mAb including detailed immunological monitoring before and following mAb infusion. Tixagevimab/Cilgavimab was developed as pre-exposure prophylaxis for individuals at increased risk of severe COVID-19 who were unlikely to mount an adequate immune response to vaccination [13]. During the COVID-19 pandemic, it was authorized or approved for this indication in selected high-risk populations, including immunocompromised individuals, in the United States, Canada, and the European Union. With the emergence and subsequent dominance of SARS-CoV-2 variants with reduced susceptibility, its clinical use was restricted or discontinued depending on the jurisdiction. Tixagevimab/Cilgavimab has not been authorized or approved for the treatment of long COVID, chronic SARS-CoV-2 infection, or persistent circulating spike protein.

The present study explored within-patient changes in circulating spike protein, autonomic receptor autoantibodies, physical performance, and patient-reported and neuropsychological outcomes following Tixagevimab/Cilgavimab administration in immunodeficient patients with long-COVID-associated ME/CFS and detectable circulating spike protein.

## 2. Methods

### 2.1. Study population and design

This prospective, non-randomized observational study initially enrolled ten immunodeficient patients with long-COVID-triggered or -aggravated ME/CFS and detectable circulating spike protein. Tixagevimab/Cilgavimab was administered for SARS-CoV-2 pre-exposure prophylaxis because of humoral and/or cellular immunodeficiency and was not assigned by the study as a treatment for long COVID. One patient withdrew consent after administration and was excluded from the analyses, leaving nine patients with paired baseline and 30-day follow-up data. Eleven healthy individuals with previous SARS-CoV-2 infection who had not developed long COVID were recruited from a separate study and provided descriptive reference data. These participants differed from the Tixagevimab/Cilgavimab-exposed cohort with respect to immune status, disease status, baseline autoantibody concentrations, and availability of clinical outcome data, as seen in Table 2. Therefore, this cohort was not considered an efficacy control group and was not used to attribute within-patient changes to Tixagevimab/Cilgavimab.

The study was approved by the ethical committee of Ruhr-University Bochum (ID 22–7491) and all participants provided written informed consent.

### 2.2. Tixagevimab/Cilgavimab administration

Passive immunization was conducted by application of Tixagevimab/Cilgavimab 150/150 mg (EVUSHELD® by AstraZeneca) between March 19th and May 21st of 2024. Tixagevimab/Cilgavimab is a combined mAb. Each agent targets the spike protein of SARS-CoV-2 and is delivered in a separate vial. Each vial needs to be filled into separate syringes to then again be injected separately intramuscular after local disinfection.

### 2.3. Spike protein elisa

Blood samples were collected into a serum collection tube 461 by Sarstedt (Germany, Cat.-No. 04.1935) and stored at −20 °C until further processing. Spike protein concentration was measured using the Human SARS-CoV-2 RBD ELISA Kit by Invitrogen (Thermo Fisher, USA, Cat.-No. EH492RB) according to the manufacturer's manual. The plates' optical density was determined using an Asys UVM 340 microplate reader (Biochrom GmbH, Germany).

### 2.4. Measurement of autonomic nervous system receptor (ANSR) auto-antibodies

Blood samples were collected into a serum collection tube by Sarstedt (Germany, Cat.-No. 04.1935) and stored at −20 °C until further processing. All ELISA systems for detection of autoantibodies were obtained from CellTrend®, Germany. β−1 Adrenergic Receptor (β1); (Cat.-No. 12,600), β−2 Adrenergic Receptor (β2); (Cat.-No. 12,700), Muscarinic Choline Receptor 3 (M3); (Cat.-No. 15,300), and Muscarinic Choline Receptor 4 (M4); (Cat.-No. 15,400) were tested. All antibodies listed above are classified as autonomic nervous system receptor (ANSR) auto-antibodies. The presence of autoantibodies was defined by using the following cut-offs: β1: >15 U/ml; β2: >8 U/ml; M3: >6 U/ml; M4: >10.7 U/ml. All analyses met the manufacturer's instructions and standards.

### 2.5. Flow cytometry

EDTA-treated whole blood was stained with optimal concentrations of each antibody for 10 min at room temperature in the dark. Samples for B cell subset analysis were washed twice with PBS (PAN-Biotech, Germany; P04–36,500) prior to antibody staining. Erythrocytes were lysed using VersaLyse (Beckman Coulter, USA; A09777) supplemented with 2.5% IOTest 3 Fixative Solution (Beckman Coulter, USA; A07800) for 30 min at room temperature in the dark. Samples for general immunophenotyping were acquired immediately, whereas samples for T and B cell subset analysis were washed twice with PBS/BSA before acquisition. All samples were analyzed using a CytoFlex flow cytometer (Beckman Coulter, USA, B53002). Absolute cell counts were determined with the Sysmex XN-330 hematology analyzer (Sysmex, Germany). These values were then combined with flow cytometry-derived percentages to calculate the absolute numbers of lymphocyte subpopulations. Quality control was performed daily using CytoFlex Daily QC Fluorospheres (Beckman Coulter, USA; C65719), according to the manufacturer's instructions. No adjustments to the compensation matrices were required throughout the study. Our gating strategies for basic immune cell phenotyping, T-cell phenotyping, activated T-cell phenotyping, B-cell phenotyping are displayed in supplemental Figures 1–4. Details about antibody analyses are provided in the Supplements.

### 2.6. Measurement of handgrip strength

As an objective parameter hand grip strength was tested using a

hydraulic hand dynamometer by Saehan® (Germany). Patients were instructed in advance. While seated, their forearm was placed on a table right in front of the patient. The hand dynamometer was then pulled for 3 s followed by a 5 s pause. This procedure was repeated ten times. After one hour the entire process was repeated.

### 2.7. ME/CFS and neuropsychological evaluation

The short-form 36 (SF-36) evaluates health related quality of life and is used in numerous studies with long-COVID patients [14–16]. In order to quantify the severity of mental and physical fatigue, all patients completed the Chalder Fatigue Scale. The Bell disability scale was also used to determine physical function and impairment. Neuropsychological assessments were conducted using the "Consortium to Establish a Registry for Alzheimer's Disease Neuropsychological Assessment Battery" (CERAD Plus). This includes testing for verbal fluency, the Boston Naming Test (modified), Mini-Mental State examination, verbal memory, constructional practice, delayed recall, processing speed using the Trail Making Test A and executive functioning using the Trail Making Test B. ME/CFS diagnosis was confirmed by the Canadian consensus criteria for all patients. For a better group description, we presented the suspected triggers of ME/CFS in Table 1.

### 2.8. Statistical analysis

Data were checked for normal distribution by the Kolmogorov-Smirnov Test. Normally distributed data were presented as mean ± standard deviation and were compared from baseline to follow-up by a paired two-tailed *t*-test. Categorical data and non-normally distributed numeric data are presented as median and interquartile range (IQR) and were compared from baseline to follow-up by a Wilcoxon-Test. P values below 0.05 were considered significant. P values were not corrected for multiple testing as this study was of exploratory nature. The normalization standard for the ELISA results was calculated using a four-parameter logistic regression. Within-patient changes were calculated as follow-up minus baseline values. Associations between changes in circulating spike protein concentrations and changes in β1-adrenergic, β2-adrenergic, M3-muscarinic, and M4-muscarinic receptor autoantibody concentrations were assessed using two-sided Spearman rank correlations. Given the small sample size and exploratory nature of these analyses, p values were not adjusted for multiple comparisons and should be interpreted descriptively. All statistical analysis was done using GraphPad Prism 10 (GraphPad Software Inc., Boston, Massachusetts, USA).

**Table 1**
Epidemiological characteristics and comorbidities of the healthy reference cohort and the Tixagevimab/Cilgavimab-exposed cohort. BMI - body mass index; COPD - chronic obstructive pulmonary disease; ESRD - end stage renal disease.

| | Reference cohort | Exposed cohort |
|---|---|---|
| Patients | **n = 11** | **n = 9** |
| Age [years] | 40±17 | 46±10 |
| Sex female | n = 4 (36%) | n = 3 (33%) |
| BMI [kg/m$^2$] | 24.6 ± 2.9 | 25.9 ± 4.3 |
| ME/CFS | | |
| Trigger SARS-CoV-2 Infection | — | n = 4 (44%) |
| Trigger Vaccination for SARS-CoV-2 | — | n = 2 (22%) |
| Another trigger worsened by SARS-CoV-2 Infection | — | n = 3 (33%) |
| Comorbidities | | |
| Hypertension | n = 3 (27%) | n = 3 (33%) |
| Hyperlipidemia | n = 0 (0%) | n = 3 (33%) |
| COPD | n = 0 (0%) | n = 0 (0%) |
| Nicotine abuse | n = 1 (9%) | n = 1 (11%) |
| ESRD | n = 0 (0%) | n = 0 (0%) |
| Diabetes mellitus | n = 0 (0%) | n = 1 (11%) |

## 3. Results

### 3.1. Patients demographic characteristics

Cohort characteristics are presented in Table 1. Ten immunodeficient patients received Tixagevimab/Cilgavimab for SARS-CoV-2 pre-exposure prophylaxis. One patient subsequently withdrew consent and was excluded from all analyses, leaving nine patients with paired baseline and 30-day follow-up data. Passive immunization was realized by Tixagevimab/Cilgavimab mAb as depicted above. All included patients presented with SARS-CoV-2 associated symptoms of ME/CFS and showed detection of spike protein in peripheral blood. Before and 30 days after passive immunization all patients underwent blood sampling, measurements of hand grip strength and various assessments for mental and physical health status.

33% of these patients were female and mean age was 46±10 years. Mean BMI was 25.9 ± 4.3 kg/m$^2$. In 67% of the patients ME/CFS symptoms were triggered by a SARS-CoV-2 infection/vaccination, while 33% reported pre-existing ME/CFS symptoms being aggravated by COVID-19. 33% had reported hypertension and hyperlipidemia, 11% confirmed nicotine abuse and diabetes mellitus. None of the patients had COPD or end-stage renal disease. In the healthy reference cohort, 36% were female, with a mean age of 40 ± 17 years and a mean body mass index (BMI) of 24.6 ± 2.9 kg/m$^2$. There were no statistically significant differences in this demographic data between patients and healthy reference cohorts.

### 3.2. Analyses of immunocompetence in patients with long-covid

To identify Long-COVID patients at risk of SARS-CoV-2 reinfection, we assessed their level of immunodeficiency by analyses of the cellular and the humoral immunity. Detailed results are displayed in supplemental Table 1. In detail, while no alterations in Ig main classes and IgG subclasses were detected, the analyses of lymphocyte subpopulations revealed reduced total lymphocyte counts, decreased numbers of CD8$^+$ T cells and CD4$^+$ central memory T cells, as well as a marked reduction in circulating plasmablasts. Furthermore, none of the long-COVID patients with the observed cellular immunodeficiency showed measurable SARS-CoV-2 spike protein-specific antibodies prior the treatment/at the baseline. These data indicated a certain degree of cellular and specific humoral deficiency and provided a rationale for the passive immunization with Tixagevimab/Cilgavimab.

### 3.3. Within-patient changes in circulating spike protein and autonomic receptor autoantibodies following Tixagevimab/Cilgavimab administration

Individual decreases in spike protein concentration were observed in five of nine patients (Fig. 1). Across the entire Tixagevimab/Cilgavimab-exposed cohort, however, median spike protein concentrations changed from 63 pg/ml (IQR 9–127) at baseline to 26 pg/ml (IQR 10–120) at follow-up, without a statistically significant within-patient change ($p = 0.63$). Thus, no group-level evidence for a reduction in circulating spike protein was observed.

Similar to spike protein, β−2-adrenoreceptor autoantibodies, muscarinic M3-receptors and muscarinic M4-receptors autoantibodies did not show significant changes from baseline to follow up (22.4 ± 8.5 vs. 18.9 ± 5.2 U/ml, $p = 0.11$; 7.4 ± 1.7 vs. 8.5 ± 5.3 U/ml, $p = 0.49$ and 10.2 ± 3.0 vs. 11.7 ± 9.4 U/ml, $p = 0.60$). In contrast, β1-adrenergic receptor autoantibody concentrations decreased from 26.8 ± 18.2 to 21.2 ± 14.7 U/ml (unadjusted $p = 0.003$), as displayed in Fig. 1.

For descriptive context, the healthy reference cohort had lower baseline concentrations of β1-adrenergic receptor autoantibodies (9.8 ± 8.3 vs. 26.8 ± 18.2 U/ml; $p = 0.0122$), β2-adrenergic receptor autoantibodies (9.1 ± 8.1 vs. 22.4 ± 8.5 U/ml; $p < 0.0001$), and M4-muscarinic receptor autoantibodies (2.9 ± 1.5 vs. 10.2 ± 3.0 U/ml;

**Table 2**
Clinical and laboratory parameters at baseline and after 30 days in the healthy reference cohort and the Tixagevimab/Cilgavimab-exposed cohort. Spike protein concentration including limit of detection (LOD), G-Protein coupled receptor autoantibodies (GPCR-Ab) concentrations including upper limit of normal (ULN), handgrip strength, Short-Form (SF-36) total score and parameters, Consortium to Establish a Registry for Alzheimer's Disease Neuropsychological Assessment Battery (CERAD-Plus) total score and its parameters, Chalder Fatigue Scale and Bell Score from baseline to follow-up. Within-patient baseline-to-follow-up comparisons were performed in the Tixagevimab/Cilgavimab-exposed cohort; $p < 0.05$ was considered statistically significant. Data in front of * showed a significant difference when comparing the healthy reference cohort vs. the intervention at time of inclusion (baseline). Significant results are printed in bold.

| Tests and Parameters | Reference cohort | | Exposed cohort | | p |
|---|---|---|---|---|---|
| | Baseline | Follow-Up | Baseline | Follow-Up | |
| Spike protein [pg/ml] (LOD <6.14 pg/ml) | — | — | 63 (9–127) | 26 (10–120) | 0.63 |
| GPCR-Ab [U/ml] | | | | | |
| β1-adrenoreceptor antibodies (ULN 15U/ml) | 9.8 ± 8.3 * | 12.7 ± 9.8 | 26.8 ± 18.2 | 21.2 ± 14.7 | **0.0030** |
| β2-adrenoreceptor antibodies (ULN 8 U/ml) | 9.1 ± 8.1 * | 10.3 ± 9.2 | 22.4 ± 8.5 | 18.9 ± 5.2 | 0.11 |
| Muscarinic M3-receptor antibodies (ULN 6 U/ml) | 6.4 ± 3.6 | 6.1 ± 4.0 | 7.4 ± 1.7 | 8.5 ± 5.3 | 0.49 |
| Muscarinic M4-receptor antibodies (ULN 10.7 U/ml) | 2.9 ± 1.5 * | 6.9 ± 5.9 | 10.2 ± 3.0 | 11.7 ± 9.4 | 0.60 |
| Handgrip strength (max.) [kg] | | | | | |
| Try 1 | — | — | 42.3 ± 17.5 | 44.7 ± 18.5 | 0.27 |
| Try 2 | — | — | 42.7 ± 19.8 | 44.0 ± 18.9 | 0.20 |
| Self-rated physical and mental health | | | | | |
| SF-36 | | | | | |
| SF-36 total score | — | — | 30.0 ± 15.2 | 33.3 ± 18.8 | 0.29 |
| Physical functioning | — | — | 32.8 ± 27.1 | 37.8 ± 29.2 | 0.22 |
| Role-physical | — | — | 0 | 11.1 ± 33.3 | 0.35 |
| Mental health | — | — | 56.9 ± 20.6 | 57.3 ± 21.4 | 0.93 |
| Vitality | — | — | 5.6 ± 7.7 | 9.4 ± 9.2 | 0.08 |
| Role-emotional | — | — | 55.6 ± 52.7 | 59.2 ± 49.4 | 0.80 |
| Social functioning | — | — | 24.8 ± 23.4 | 22.1 ± 18.4 | 0.69 |
| General health | — | — | 31.0 ± 21.3 | 23.9 ± 13.6 | 0.11 |
| Bodily pain | — | — | 43.1 ± 29.6 | 42.7 ± 30.3 | 0.95 |
| CERAD-Plus | | | | | |
| Total | 146.0 ± 12.4 | 146.9 ± 10.4 | 136.3 ± 18.8 | 140.3 ± 19.9 | 0.27 |
| Verbal fluency | 22.4 ± 6.3 | 22.9 ± 5.6 | 24.2 ± 7.3 | 25.4 ± 8.5 | 0.36 |
| Boston Naming Test | 15.0 ± 0 | 15.0 ± 0 | 15±0 | 15±0 | 1.0 |
| Mini Mental Status | 30.0 ± 0 | 30.0 ± 0 | 30±0 | 30±0 | 1.0 |
| Memorizing list of word | 25.6 ± 2.1 | 26.3 ± 2.5 | 21.3 ± 3.7 | 22.3 ± 6.5 | 0.49 |
| Reproducing words | 7.4 ± 1.1 | 9.6 ± 0.7 | 8.1 ± 1.6 | 8.1 ± 2.6 | 0.99 |
| Intrusions | 0.1 ± 0.3 | 0.0 ± 0 | 0.1 ± 0.3 | 0.3 ± 0.7 | 0.45 |
| Copying figures | 11.0 ± 0 | 11.0 ± 0 | 10.0 ± 1.3 | 10.3 ± 1.1 | 0.35 |
| Memorizing figures | 11.0 ± 0 | 11.0 ± 0 | 10.1 ± 1.3 | 10.1 ± 1.3 | 0.99 |
| Phonemic fluency | 21.8 ± 4.7 | 21.2 ± 4.2 | 17.7 ± 6.8 | 17.6 ± 6.9 | 0.95 |
| Trail making Test A | 21.0 ± 4.0 | 20.5 ± 4.0 | 36.3 ± 14.5 | 33.3 ± 13.3 | 0.11 |
| Trail making Test B | 45.3 ± 12.5 | 43.0 ± 14.1 | 83.3 ± 48.8 | 65.4 ± 23.0 | 0.21 |
| Other questionnaires | | | | | |
| Chalder Fatigue Scale | — | | 9.9 ± 1.8 | 9.9 ± 1.7 | 0.99 |
| Bell Score | — | | 31.1 ± 9.3 | 35.6 ± 15.1 | 0.10 |

$p < 0.0001$). Because the cohorts differed in immune and disease status, these baseline comparisons are descriptive and were not used to assess treatment efficacy.

### 3.4. Within-patient changes in clinical outcomes and handgrip strength

None of the assessed clinical outcomes changed significantly between baseline and 30-day follow-up. The SF-36 total score changed from 30.0 ± 15.2 to 33.3 ± 18.8 ($p = 0.29$), and the CERAD-Plus total score changed from 136.3 ± 18.8 to 140.3 ± 19.9 ($p = 0.27$; Fig. 2). No significant changes were observed in Bell score, Chalder Fatigue Scale, or maximum handgrip strength. Although individual participants showed numerical increases in some outcomes, these observations do not provide evidence of a clinical treatment effect.

### 3.5. Correlation analyses of collected clinical and laboratory data

The changes analyzed here represent paired within-patient differences between measurements obtained immediately before Tixagevimab/Cilgavimab administration and measurements obtained 30 days after administration. Exploratory Spearman correlation analyses did not identify a statistically significant association between the change in circulating spike protein concentration and changes in any of the assessed autonomic receptor autoantibodies. The correlation was moderate and positive for β1-adrenergic receptor autoantibodies ($\rho = 0.52$, exact $p = 0.162$), indicating that larger decreases in spike protein tended to be accompanied by larger decreases in β1-adrenergic receptor autoantibody concentrations. However, this association did not reach statistical significance. No relevant associations were observed for β2-adrenergic receptor ($\rho = -0.10$, exact $p = 0.810$), M3-muscarinic receptor ($\rho = 0.17$, exact $p = 0.678$), or M4-muscarinic receptor autoantibodies ($\rho = -0.13$, exact $p = 0.744$). Given the sample size of nine patients, these exploratory analyses had limited statistical power.

## 4. Discussion

There is an urgent clinical need for evidence-based therapies of long-COVID. The identification of promising candidate approaches necessitates preliminary exploratory studies. Recently, Scheppke et al. published three cases of patients with long-COVID that received an infusion with Casirivimab/Imdevimab, two spike protein-targeting antibodies. All three patients experienced remission of their disabling disease and rapidly returned to previous physical and mental wellbeing [17]. The persistence of spike protein is broadly discussed as a potential cause for long-COVID. Swank et al. found a spike protein persistence in 60% of long-COVID patients up to 12 months after diagnosis, whereas 0% of

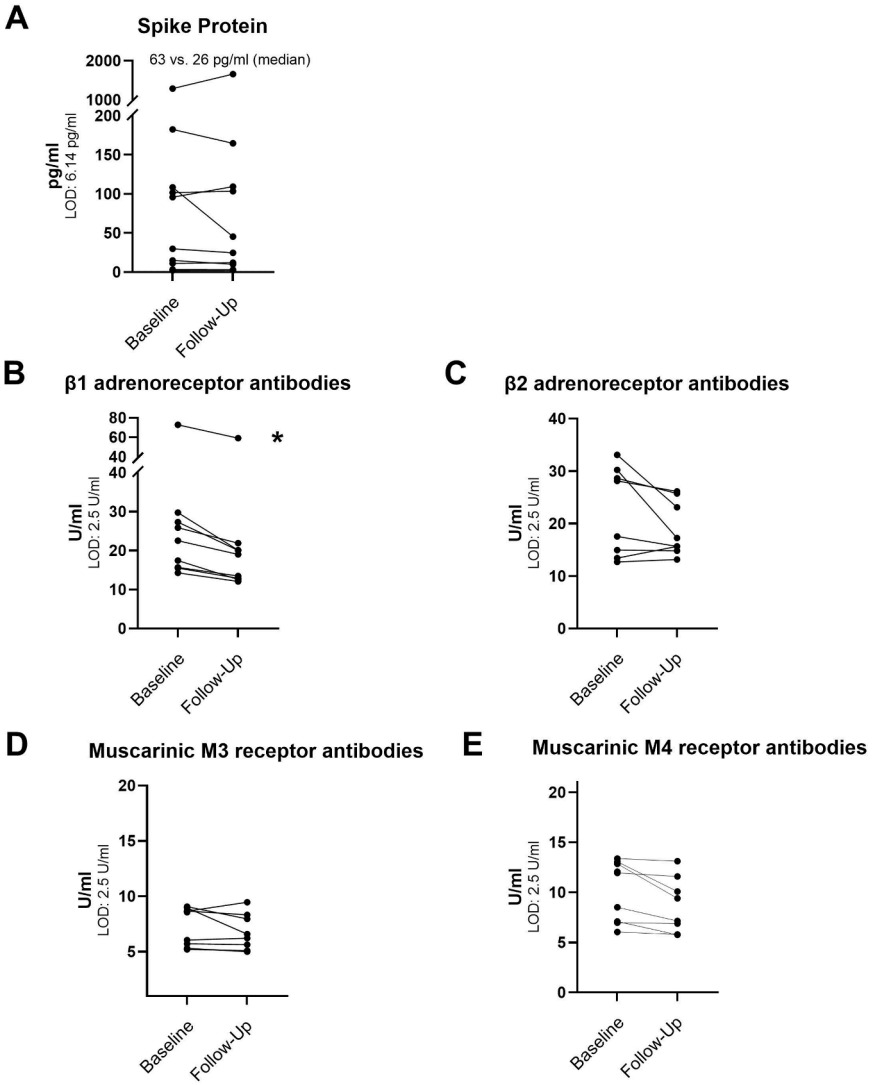



**Fig. 1.** Spike protein concentration (A), Beta-1 and −2 receptor antibody concentration (B and C) and muscarinic M3 and M4 receptor antibody concentration (D and E) for the exposed cohort at baseline and after 30 days. Figures A-E include 9 patients. Significant results are marked as *. $P < 0.05$ was considered significant.

patients after SARS-CoV-2 infection without long-COVID showed spike persistence [18]. Spike protein persistence is supposed to be associated with a pro-inflammatory effect either by inflammation of the endothelium, perivascular inflammation of the brain or by autoimmunity due to similar epitopes with human chaperones [19–21]. Despite these hypotheses, the exact role of spike protein persistence in the context of long-COVID in SARS-CoV-2 induced ME/CFS remains unclear.

The aim of the present study was to explore whether Tixagevimab/Cilgavimab administration was followed by within-patient changes in circulating spike protein, autonomic receptor autoantibodies, and clinical outcomes. Applying mAbs against spike protein should reduce spike protein induced immune activation and minimize molecular mimicry-driven autoimmunity. Searching for an approved and an available mAb for the application in the clinical routine, we identified Tixagevimab/Cilgavimab approved for the preexposure prophylaxis in patients at risk for COVID-19. Tixagevimab/Cilgavimab is a combination of extended-half-life neutralizing mAbs that bind to distinct epitopes on the receptor binding domain (RBD) of the SARS-CoV-2 spike protein and demonstrates a maintained neutralization activity against all omicron subvariants up to and including BA.5 [22].

In our small pilot trial, we explored changes following Tixagevimab/Cilgavimab administration applied in patients with immunodeficiency demonstrating symptoms of long-COVID associated ME/CFS and detectable spike protein in peripheral blood. According to the time points of SARS-CoV-2 infection, all patients were infected by omicron variants known to circulate since November 2021 [22].

In this exploratory cohort, Tixagevimab/Cilgavimab administration was followed by a nominally significant within-patient reduction in β1-adrenergic receptor autoantibody concentrations. In contrast, no statistically significant within-patient changes were observed in circulating spike protein concentrations, the other assessed autonomic receptor autoantibodies, handgrip strength, or patient-reported and neuropsychological outcomes. Although individual patients showed numerical decreases in spike protein concentrations or numerical improvements in some clinical outcomes, these observations do not provide evidence of a clinical treatment effect. Given the non-randomized observational design and the absence of an untreated disease-matched control group, the observed changes cannot be causally attributed to Tixagevimab/Cilgavimab. Changes in circulating spike protein concentrations were not significantly correlated with changes in any of the assessed autonomic receptor autoantibodies. A moderate but non-significant positive association was observed between changes in spike protein and β1-adrenergic receptor autoantibody concentrations. This direction of association would be compatible with the proposed hypothesis that a reduction in persistent spike protein might be accompanied by reduced autoreactivity. However, the association did not reach statistical

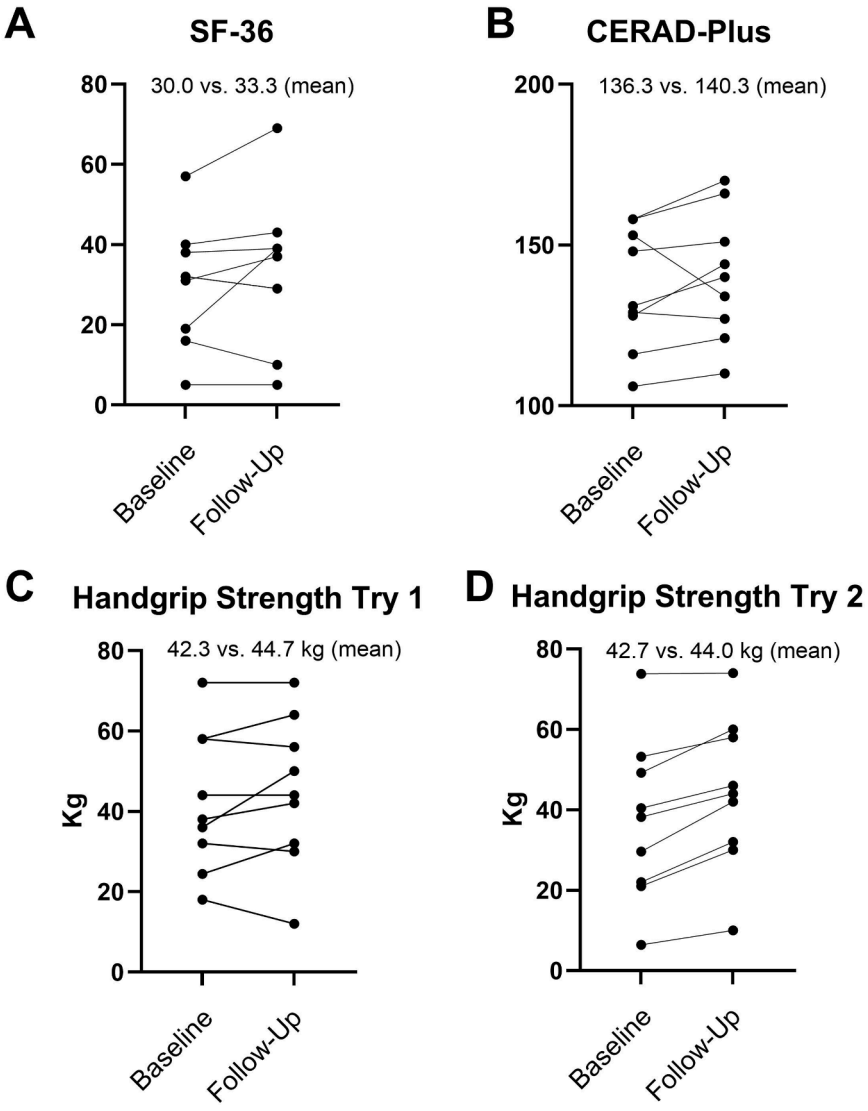



**Fig. 2.** Short-Form 36 total score (SF-36) (A), Consortium to Establish a Registry for Alzheimer's Disease Neuropsychological Assessment Battery (CERAD-Plus) (B) and maximum handgrip strength for try 1 (C) and try 2 (D before (baseline) and 30 days after administration of Tixagevimab/Cilgavimab (follow-up). Figures A-D include 9 patients. Significant results are marked as *. $P < 0.05$ was considered significant.

significance, and the small sample size precludes mechanistic conclusions. Thus, the present data neither confirm nor reliably exclude a relationship between spike protein reduction and changes in β1-adrenergic receptor autoantibodies.

The healthy reference cohort showed lower baseline concentrations of several autonomic receptor autoantibodies than the Tixagevimab/Cilgavimab-exposed cohort. However, these differences should not be interpreted as evidence of a treatment effect because the cohorts differed fundamentally in immune status and disease status. Similar findings have previously been reported by our group and others when comparing patients with long-COVID to healthy controls [3]. Furthermore, we observed elevated IgE levels in the healthy reference group, which may reflect the increased risk of allergic diseases in long-COVID described by Oh et al. [23]. However, baseline data on the prevalence of allergic diseases at inclusion are not available.

As described above, we and others have demonstrated that levels of autonomic nervous system receptor autoantibodies are associated with the severity of neurological symptoms in long-COVID [3]. Beyond long-COVID, these and other GPCR-autoantibodies have been described in diseases like the complex regional pain syndrome and systemic sclerosis [24–26]. Since autoantibodies constitute a promising therapeutic target for patients with long-COVID, there have been different recent approaches either suppressing antibody formation via immunosuppression or eliminating existing antibodies via plasma exchange or immunoadsorption [5,27].

Although individual biomarker and symptom changes were observed following Tixagevimab/Cilgavimab administration, the present study was not designed to demonstrate therapeutic efficacy. Spike protein concentrations decreased in five of nine patients. The absence of a decrease in the remaining four patients might partly reflect infection with variants showing reduced susceptibility to Tixagevimab/Cilgavimab; however, variant identity was not available, and this explanation remains speculative.

β1-adrenergic receptors are G-protein-coupled receptors and represent the predominant β-adrenergic receptor subtype in the human heart, mediating positive chronotropic and inotropic responses to catecholamines [28]. In myocarditis and dilated cardiomyopathy, β1-adrenergic receptor autoantibodies have been shown to exert β1-agonist-like effects in cardiomyocyte bioassays, including a sustained increase in beating frequency [29]. However, the receptor-coated ELISA used in the present study quantifies IgG binding, with signal intensity reflecting antibody concentration and/or avidity, but does not determine whether the

detected autoantibodies exert agonistic, antagonistic, or functionally neutral effects [30]. Autoantibodies targeting vasoregulatory and autonomic nervous system receptors have been detected in long COVID using both binding ELISAs and functional bioassays [3,24]. In our previous ELISA-based study, higher autoantibody concentrations were associated with greater cognitive and physical impairment [3]. However, these observational findings do not establish causality, and clinically validated receptor-specific thresholds for predicting symptom severity or treatment response are lacking. The reduction in β1-adrenergic receptor autoantibody concentrations observed in the present study may therefore indicate a change in binding autoreactivity, but it cannot be interpreted as evidence of altered receptor function or as a surrogate for clinical benefit. Furthermore, the mean concentration at follow-up remained above the manufacturer-defined positivity cut-off of 15 U/ml.

The strength of the present study is the broad variety of questionnaires used to describe symptoms of ME/CFS. The different questionnaires thereby encompass the broad variety of symptoms and their effects on quality of life in this entity. In addition to these questionnaires, hand grip strength serves as a more objective measure of physical performance. Another strength is the strict inclusion criterion of ME/CFS as defined by the Canadian Criteria. This fact is of importance, since long-COVID is a highly heterogeneous disease. Patients complain about a broad spectrum of symptoms, ranging from purely somatic symptoms due to direct organ damage to fatigue, a symptom that can barely be assessed objectively.

This exploratory study has several important limitations. First, the sample size was small, and the study had a non-randomized observational design. Tixagevimab/Cilgavimab was administered for SARS-CoV-2 pre-exposure prophylaxis rather than assigned by the study as a treatment for long COVID. Consequently, the observed changes cannot be causally attributed to Tixagevimab/Cilgavimab. Second, the healthy reference cohort differed from the exposed cohort in immune status, disease status, baseline autoantibody concentrations, and availability of clinical outcome data and therefore cannot be considered a valid efficacy control group. An untreated disease-matched cohort of immunodeficient patients with long COVID would have been required to distinguish administration-associated changes from natural disease fluctuation, regression to the mean, or placebo effects. Third, the follow-up period was limited to 30 days, which precluded assessment of the durability of the observed biomarker changes and the possibility of delayed clinical effects. Fourth, the study included only immunodeficient patients, substantially limiting the generalizability of the findings to the broader long-COVID population. We had also intended to include a group of patients with ME/CFS without previous SARS-CoV-2 infection. However, the regional SARS-CoV-2 seroprevalence was estimated to exceed 90%, making it practically impossible to recruit such a cohort. Longer-term follow-up and adequately powered randomized studies using an untreated disease-matched control group are therefore required.

Furthermore, Tixagevimab/Cilgavimab is intentionally engineered to minimize activation of the immune system. By blocking SARS-CoV-2 entry into host cells, it neutralizes the virus without eliciting a strong inflammatory response [31]. However, this attenuated effector function may limit its efficacy in the context of long-COVID, particularly in cases involving spike protein persistence in various tissues, where a more robust inflammatory response might be necessary for effective antigen clearance. In contrast, agents with effector function activity, such as intravenous immunoglobulins (IVIG), may offer greater therapeutic potential in long-COVID. Current data, however, remain limited and heterogeneous. Morse et al. provide a comprehensive overview of IVIG's immunomodulatory mechanisms and highlight early yet promising findings. In a case series, Thompson et al. reported significant symptomatic improvements in six out of nine patients with long-COVID following long-term administration of high-dose IVIG [32,33]. Ongoing clinical trials, including the RECOVER-AUTONOMIC study (comparing IVIG to placebo; Clinicaltrials.gov ID NCT06305780) and the IN-PASCI trial (comparing IVIG with placebo with a crossover and wash out design, Clinicaltrials.gov ID NCT05350774), aim to generate more robust evidence regarding IVIG's efficacy in this setting. Additional therapeutic strategies under investigation include immunomodulators and anti-inflammatory agents (e.g., Baricitinib), antiviral therapies (e.g., Nirmatrelvir/Ritonavir [Paxlovid]), monoclonal antibodies (e.g., Sipavibart), and alternative approaches such as low-dose naltrexone. Beyond Tixagevimab/Cilgavimab, other monoclonal antibodies assessed for the prevention of SARS-CoV-2 infection include Casirivimab/Imdevimab in pre- and post-exposure prophylaxis, Adintrevimab, and Pemivibart in pre-exposure prophylaxis [34–37]. All of these agents demonstrated efficacy in preventing SARS-CoV-2 infection in their respective trials. However, none of these studies evaluated outcomes related to the prevention or treatment of long-COVID. To date, the only published case series examining monoclonal antibodies for long-COVID was conducted by Scheppke et al., who reported reduction of symptoms in all three patients treated with Casirivimab/Imdevimab [17].

In conclusion, Tixagevimab/Cilgavimab administration was followed by a nominally significant within-patient reduction in β1-adrenergic receptor autoantibody concentrations, but no conclusive reduction in circulating spike protein or improvement in clinical outcomes was demonstrated. These findings should be regarded as preliminary and hypothesis-generating. Their interpretation is limited by the small sample size, non-randomized observational design, brief 30-day follow-up period, inclusion of only immunodeficient patients, and absence of an untreated disease-matched control group. Consequently, the present study does not establish therapeutic efficacy or a causal relationship between Tixagevimab/Cilgavimab administration and the observed biomarker change. Adequately powered randomized trials using a disease-matched control group and longer follow-up are required to evaluate further clinical and immunological effects.

## Abbreviations

Ab (Antibodies).
ANSR (Autonomic Nervous System Receptor).
BMI (Body Mass Index).
CCC (Canadian Consensus Criteria).
CERAD (Consortium to Establish a Registry for Alzheimer's Disease).
COPD (Chronic Obstructive Pulmonary Disease).
ELISA (Enzyme-Linked Immunosorbent Assay).
ESRD (End Stage Renal Disease).
Ig (Immunoglobulin).
IQR (Interquartile Range). mAb (Monoclonal Antibody).
ME/CFS (Myalgic Encephalomyelitis / Chronic Fatigue Syndrome).
RBD (Receptor Binding Domain).
SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2).
SF-36 (Short Form 36).
WHO (World Health Organization).

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work in parts of the literature search and evidence synthesis the authors were assisted by OpenEvidence (OpenEvidence.ai), an AI-based research tool used to identify and summarize peer-reviewed studies. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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## CRediT authorship contribution statement

**Maximilian Seidel:** Writing – original draft, Formal analysis. **Moritz Anft:** Methodology, Data curation, Conceptualization. **Lea Wiemers:** Project administration, Data curation. **Felix S. Seibert:** Validation, Supervision. **Sebastian Bertram:** Supervision, Formal analysis. **Kamil Rosiewicz:** Visualization, Formal analysis. **Ulrik Stervbo:** Visualization, Software. **Timm H. Westhoff:** Writing – review & editing, Conceptualization. **Nina Babel:** Writing – review & editing, Conceptualization.

## Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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## Supplementary materials

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